

# AI Critique

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Overall, the AI tool demonstrated strong capabilities in generating syntactically valid JMeter XML test plans (.jmx) and extracting baseline execution metrics from raw .jtl logs; however, a detailed technical audit across Load, Stress, Spike, and Endurance scenarios revealed critical flaws in load modeling, listener scoping, hardware context awareness, and metric interpretation. Specifically, the AI failed to account for single-node development constraints (Node.js single-threaded event loop and SQLite single-writer lock architecture), proposing unrealistic workload configurations (100–500 VUs). Furthermore, the AI introduced severe memory management risks by placing heavy listeners (View Results Tree) directly on high-concurrency CLI runs, misconfigured global Authorization headers, and hallucinated architectural fixes (such as offloading non-existent bcrypt CPU rounds or forcing PostgreSQL migrations). The AI also misattributed steady-state think-time throughput plateaus (~125 RPS) as hardware ceilings, ignoring the system's empirical capacity to absorb 274+ RPS bursts. Ultimately, human QA intervention was indispensable to downscale thread distributions, enforce proper JMeter scoping, apply SQLite WAL mode optimizations, and identify the true 500 ms event-loop stall during prolonged soak testing, proving that AI generation must be paired with rigorous human domain expertise to deliver meaningful performance engineering.